

gPXE: A Troubleshooting Note for Sysadmins



Preboot eXecution Environment (PXE) provides you the ability to boot computers using the network interface. gPXE is an open source PXE implementation and network boot loader. It replaces proprietary PXE ROMs and has many functionalities, such as retrieving data through protocols like HTTP, iSCSI, etc. Some of the latest operating systems and hypervisors like VMware ESXi 5.x need gPXE as a pre-requisite for deploying via the network. If you already have a legacy PXE implementation, you can migrate to gPXE by placing the gPXE executable on your TFTP server. The PXE-enabled machines download gPXE via TFTP and instantly become gPXE-enabled.

Setting up gPXE

Read the white paper posted in the *Dell Tech Center* to set up gPXE for network deployment of operating systems. Though it talks about setting up gPXE specifically for VMware ESXi 5, it is also useful for setting up the gPXE environment and thus meeting the pre-requisites for deploying various other operating systems over the network.

Getting all set to boot from gPXE

While booting servers to PXE and then gPXE, some NICs may fail with an error, as shown in Figure 1.

You might wonder why you need to again request for a DHCP IP, since it has already got one at the beginning? The reason is that when the chain loaded gPXE starts up (the PXE-enabled NICs download gPXE via TFTP), it issues a fresh DHCP request because it does not see the previous DHCP IP issued for the legacy PXE. The DHCP connection time-out is due to a timing issue with respect to the NIC firmware when the chain load of gPXE occurs.

Workarounds

There are a few workarounds to this problem, which are discussed below.

Modifying the PXE-chainloadable gPXE image: This workaround is to recreate the gPXE image with a custom script, which is nothing but a sleep before the NIC queries for the DHCP IP again after the gPXE chain load. The detailed